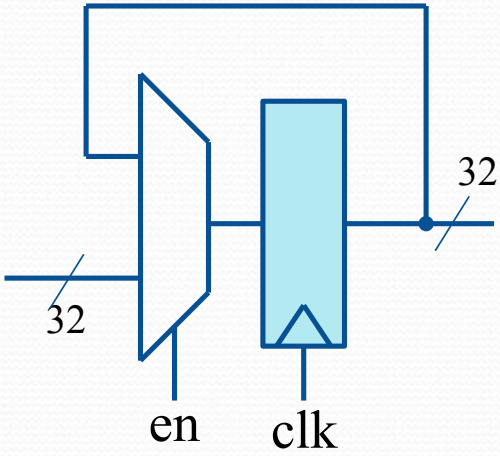
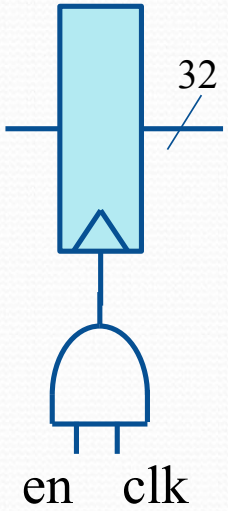
Exercise 16 Gated Clocks:

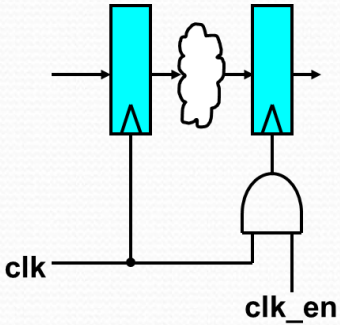




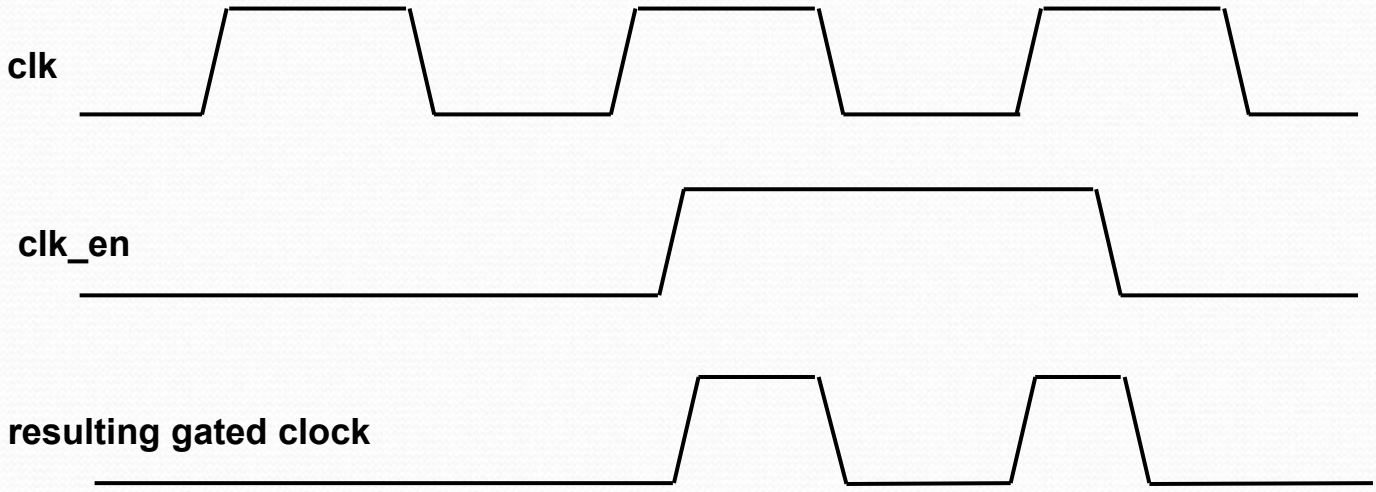
Recirculating seems rather wasteful we need a 2:1 mux for each flop. Plus all the cap and gates inside the flops on the clock network will be toggling.

Wouldn't it be better to just gate the clock? Pretty much same functionality but should save area and power? Yes, but you have to do it right.

Exercise 16 Gated Clocks:

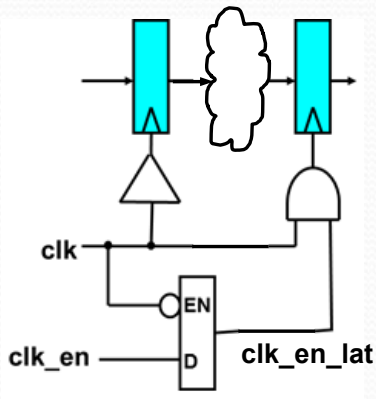


Something this simple does not work. Recall the enable function is probably coming from a SM. So it changes slightly after the rising edge of clock.



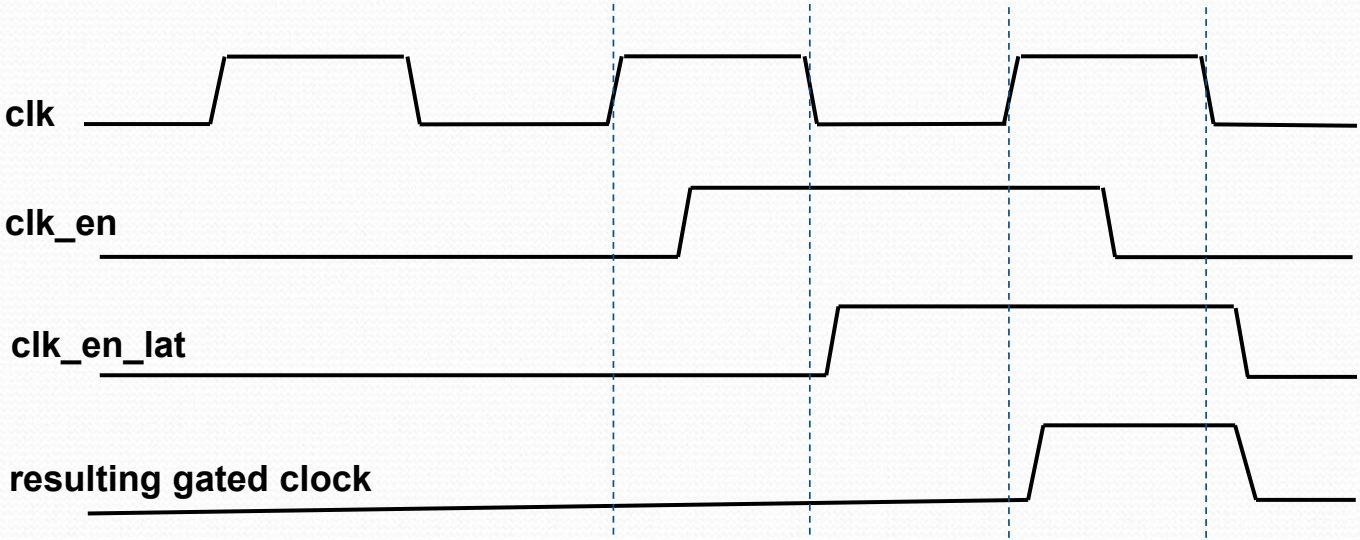
That's pretty messed up!

Exercise 16 Gated Clocks:



Have to use an enable low latch to latch the clock enable signal. This ensures the enable is setup and ready for the next high phase of clock

Have to balance from clock root to all leaf nodes of clock



Exercise 16 Gated Clocks:

- Work in groups of two
- On the website you will find two files:
  - **mult_accum.sv** (design that uses recirculating flops)
  - **mult_accum.dc** (synthesis script)
- Synthesize **mult_accum.sv** using the provided script
  - Note the area, and the power estimate
  - Note whether or not it passes timing
- Now modify the design to use gated clocks instead of recirculating.
- Modify the synthesis script as needed (minor change to read in **mult_accum_gated.sv** instead of **mult_accum.sv**)
- Again synthesize and remark on the difference in both area and timing.
- Submit the area and dynamic power numbers for both designs to the dropbox for this exercise. (both people submit)

No clock gating Area :

```
Number of ports:                          83
Number of nets:                          796
Number of cells:                         689
Number of combinational cells:           607
Number of sequential cells:               81
Number of macros/black boxes:              0
Number of buf/inv:                        84
Number of references:                     39

Combinational area:            1569.085075
Buf/Inv area:                   130.630018
Noncombinational area:          535.227282
Macro/Black Box area:             0.000000
Net Interconnect area:          356.419649

Total cell area:               2104.312356
Total area:                    2460.732005
1
```

Power:

Power Group	Internal Power	Switching Power	Leakage Power	Total Power (%)	Attrs
io_pad	0.0000	0.0000	0.0000	0.0000 (0.00%)	
memory	0.0000	0.0000	0.0000	0.0000 (0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000 (0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000 (0.00%)	
register	0.0000	0.4498	1.1407e+07	11.8565 (18.62%)	
sequential	0.0000	0.0000	0.0000	0.0000 (0.00%)	
combinational	0.0000	6.6058	4.5210e+07	51.8156 (81.38%)	
Total	0.0000 uW	7.0556 uW	5.6616e+07 pW	63.6721 uW	

1

Timing :

Point	Incr	Path
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	0.75	0.75 r
A[1] (in)	0.00	0.75 r
mult_15/A[1] (mult_accum_DW02_mult_0)	0.00	0.75 r
mult_15/U16/Y (NAND2X0_RVT)	0.04	0.79 f
mult_15/U14/Y (XOR2X2_RVT)	0.12	0.92 r
mult_15/S2_2_5/S (FADDX1_RVT)	0.16	1.08 f
mult_15/S2_3_4/S (FADDX1_RVT)	0.17	1.25 r
mult_15/S2_4_3/CO (FADDX1_RVT)	0.12	1.37 r
mult_15/S2_5_3/S (FADDX1_RVT)	0.18	1.55 f
mult_15/S2_6_2/S (FADDX1_RVT)	0.17	1.72 r
mult_15/S4_1/CO (FADDX1_RVT)	0.12	1.84 r
mult_15/U35/Y (XOR2X2_RVT)	0.14	1.97 f
mult_15/FS_1/A[7] (mult_accum_DW01_add_1)	0.00	1.97 f
mult_15/FS_1/U12/Y (AND2X1_RVT)	0.07	2.04 f
mult_15/FS_1/U48/Y (AOI21X1_RVT)	0.10	2.14 r
mult_15/FS_1/U46/Y (OA21X1_RVT)	0.07	2.21 r
mult_15/FS_1/U41/Y (OA21X1_RVT)	0.07	2.28 r
mult_15/FS_1/U39/Y (OA21X1_RVT)	0.07	2.36 r
mult_15/FS_1/U37/Y (AND2X1_RVT)	0.06	2.42 r
mult_15/FS_1/U36/Y (OA22X1_RVT)	0.08	2.50 r
mult_15/FS_1/U6/Y (XOR2X2_RVT)	0.11	2.61 f
mult_15/FS_1/SUM[13] (mult_accum_DW01_add_1)	0.00	2.61 f
mult_15/PRODUCT[15] (mult_accum_DW02_mult_0)	0.00	2.61 f
U89/Y (MUX21X2_RVT)	0.08	2.69 f
prod_reg_reg[15]/D (DFFX1_RVT)	0.01	2.70 f
data arrival time		2.70
clock clk (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
prod_reg_reg[15]/CLK (DFFX1_RVT)	0.00	2.50 r
library setup time	-0.05	2.45
data required time		2.45
data required time		2.45
data arrival time		-2.70
slack (VIOLATED)		-0.25

With clock gating Area: went down

saed32rvt_tt0p85v25c (File: /cae/apps/data/saed32

Number of ports:	83
Number of nets:	767
Number of cells:	653
Number of combinational cells:	508
Number of sequential cells:	84
Number of macros/black boxes:	0
Number of buf/inv:	76
Number of references:	46
Combinational area:	1383.559942
Buf/Inv area:	121.734978
Noncombinational area:	581.481484
Macro/Black Box area:	0.000000
Net Interconnect area:	285.114879
Total cell area:	1965.041426
Total area:	2250.156306

1

Power: went up

Power Group	Internal Power	Switching Power	Leakage Power	Total Power (%)	Attrs
io_pad	0.0000	0.0000	0.0000	0.0000 (0.00%)	
memory	0.0000	0.0000	0.0000	0.0000 (0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000 (0.00%)	
clock_network	0.0000	6.8038	3.2385e+05	7.1276 (10.78%)	
register	0.0000	0.3484	1.4014e+07	14.3621 (21.73%)	
sequential	0.0000	0.0000	0.0000	0.0000 (0.00%)	
combinational	0.0000	6.1306	3.8473e+07	44.6035 (67.49%)	
Total	0.0000 uW	13.2827 uW	5.2811e+07 pW	66.0933 uW	

1

Timing: better

```
Startpoint: en_stg2_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: clk_en_lat2_reg
          (positive level-sensitive latch clocked by clk')
Path Group: clk
Path Type: min

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
mult_accum_gated   8000                  saed32rvt_tt0p85v25c

Point                                    Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                    2.50       2.50
clock network delay (ideal)              0.00       2.50
en_stg2_reg/CLK (DFFX1_RVT)              0.00       2.50 r
en_stg2_reg/Q (DFFX1_RVT)                0.13       2.63 r
clk_en_lat2_reg/D (LATCHX1_RVT)          0.01       2.64 r
data arrival time                                   2.64

clock clk' (fall edge)                   2.50       2.50
clock network delay (ideal)              0.00       2.50
clk_en_lat2_reg/CLK (LATCHX1_RVT)        0.00       2.50 f
library hold time                       -0.04       2.46
data required time                                  2.46
-----------------------------------------------------------
data required time                                  2.46
data arrival time                                  -2.64
-----------------------------------------------------------
slack (MET)                                         0.18
```